Oxidation Number 1

Essential Pre-Uni Chemistry K1.1

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Give the oxidation number of nitrogen in the following compounds:

Part A

NH_3

NH_3

Part B

NO

NO

Part C

N_2

N_2

Part D

NO_2

NO_2

Part E

HNO_3

HNO_3

Part F

$Ca(NO_3)_2$

$Ca(NO_3)_2$

Part G

N_2H_4

N_2H_4

Part H

Mg_3N_2

Mg_3N_2

Part I

NCl_3

NCl_3

Part J

NO^+

NO^+

Question deck:

STEM SMART Chemistry Week 10

Oxidation Number 2

Essential Pre-Uni Chemistry K1.2

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Write down the oxidation number of:

Part A

H_2O

Oxygen in H_2O

Part B

H_2SO_4

Sulfur in H_2SO_4

Part C

H_3PO_4

Phosphorus in H_3PO_4

Part D

H_3PO_3

Phosphorus in H_3PO_3

Part E

ClO_2

Chlorine in ClO_2

Part F

OF_2

Oxygen in OF_2

Part G

Sodium nitrite

Nitrogen in sodium nitrite $NaNO_2$

Part H

Ammonium sulfate

Nitrogen in ammonium sulfate $(NH_4)_2(SO_4)$

Part I

Hydrogen peroxide

Oxygen in hydrogen peroxide H_2O_2

Part J

$VO_2{}^+$

V in $VO_2{}^+$

Part K

VO^{2+}

V in VO^{2+}

Part L

Hg_2^{2+}

Hg in Hg_2^{2+}

Part M

$Cr_2O_7^{2-}$

Cr in $Cr_2O_7^{2-}$

Part N

MnO_4^-

Mn in MnO_4^-

Part O

I_3^-

I in I_3^-

Question deck:

STEM SMART Chemistry Week 10

Oxidation Number 3

Essential Pre-Uni Chemistry K1.3

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Select the element or species that is being reduced in the following reactions

Part A

(a)

$CuO + H_2 \longrightarrow Cu + H_2O$

- Cu in CuO
- H in H_2
- O in CuO
- O in H_2O

Part B

(b)

$C_3H_6 + H_2 \longrightarrow C_3H_8$

- C_3H_8
- H in C_3H_6
- H in H_2
- C_3H_6

Part C

(c)

$2\,Na + Br_2 \longrightarrow 2\,NaBr$

- Na in NaBr
- Br_2
- Na
- Br in NaBr

Part D

(d)

$H_2O_2 + 2\,FeSO_4 + H_2SO_4 \longrightarrow 2\,H_2O + Fe_2(SO_4)_3$

- H_2O_2
- S in H_2SO_4
- Fe in $FeSO_4$
- S in $FeSO_4$

Part E

(e)

$ZnCl_2 \longrightarrow Zn + Cl_2$

- Zn in $ZnCl_2$
- Cl in Cl_2
- Cl in $ZnCl_2$

Part F

(f)

$Fe_2(SO_4)_3 + Zn \longrightarrow 2\,FeSO_4 + ZnSO_4$

- S in $Fe_2(SO_4)_3$
- Zn
- Fe in $Fe_2(SO_4)_3$
- Zn in $ZnSO_4$

Part G

(g)

$NiSO_4 + Fe \longrightarrow FeSO_4 + Ni$

- O in $NiSO_4$
- Ni in $NiSO_4$
- S in $NiSO_4$
- Fe

Part H

(h)

$4\,C_3H_6O + NaBH_4 + 4\,H_2O \longrightarrow 4\,C_3H_8O + NaB(OH)_4$

- B in $NaBH_4$
- H in $NaBH_4$
- H in C_3H_6O
- C_3H_6O

Question deck:

STEM SMART Chemistry Week 10

Oxidation Number 4

Essential Pre-Uni Chemistry K1.4

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Select the element or species that is being oxidised in the following reactions.

Part A

(a)

$2\,Al + Cr_2O_3 \longrightarrow Al_2O_3 + 2\,Cr$

- Al in Al_2O_3
- O in Cr_2O_3
- Al
- Cr in Cr_2O_4

Part B

(b)

$2\,NH_3 + 3\,CuO \longrightarrow N_2 + 3\,Cu + 3\,H_2O$

- N in N_2
- NH_3
- Cu in CuO
- O in CuO

Part C

(c)

$2\,Cu^{2+} + 4\,I^- \longrightarrow 2\,CuI + I_2$

- Cu in CuI
- I in CuI
- Cu^{2+}
- I^-

Part D

(d)

$6\,PbO + O_2 \longrightarrow 2\,Pb_3O_4$

- O in PbO
- Pb in PbO
- O in O_2
- Pb in Pb_3O_4

Part E

(e)

$H_2O_2 + SO_2 \longrightarrow H_2SO_4$

- SO_2
- H_2O_2
- H in H_2SO_4
- S in H_2SO_4

Part F

(f)

$3\,H_2SO_4 + 2\,NaBr \longrightarrow 2\,NaHSO_4 + Br_2 + SO_2 + 2\,H_2O$

- Na in NaBr
- H in H_2SO_4
- S in H_2SO_4
- Br in NaBr

Part G

(g)

$Mg + 2\,CH_3COOH \longrightarrow Mg(CH_3COO)_2 + H_2$

- H_2
- C in CH_3COOH
- Mg in $Mg(CH_3COO)_2$
- Mg

Part H

(h)

$2\,Fe^{3+} + 6\,ClO^- + 4\,OH^- \longrightarrow 2\,FeO_4{}^{2-} + 3\,Cl_2 + 2\,H_2O$

- Fe^{3+}
- O in ClO^-
- OH^-
- Cl in ClO^-

Question deck:

STEM SMART Chemistry Week 10

Oxidation States of S and N

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Part A

Oxidation states of sulfur

In which of the following pairs of species is the sulfur in the same oxidation state in both members of the pair?

1. SF_6 and SO_4^{2-}
2. SO_2 and HSO_3^-
3. $S_2O_3^{2-}$ and $S_4O_6^{2-}$

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct
- **3** only is correct

Part B

Oxidation states of nitrogen

In which of the following conversions does the oxidation number of the nitrogen change by two?

1. $NH_2OH \longrightarrow NH_3$
2. $N_2 \longrightarrow NO$
3. $NO_2 \longrightarrow HNO_3$

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct
- **3** only is correct

Part A adapted with permission from UCLES, A-Level Chemistry, June 1990, Paper 1, Question 35;
Part B adapted with permission from UCLES, A-Level Chemistry, June 1994, Paper 4, Question 34

Question deck:
STEM SMART Chemistry Week 10

Rocket and Metal Reduction

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Part A

Solid rocket booster

The propellant used in the solid rocket booster of a space shuttle is a mixture of aluminium and compound **X**. Compound **X** contains chlorine in an oxidation state of $+7$.

Which of the following could be compound **X**?

- NCl_3
- NH_4Cl
- NH_4ClO_3
- NH_4ClO_4
- N_2H_5Cl

Part B

Metals and reduction

In which of the following changes has the metal undergone reduction?

- $[Cr(OH)_6]^{3-} \longrightarrow CrO_4^{2-}$
- $[Co(NH_3)_6]^{2+} \longrightarrow [CoCl_4]^{2-}$
- $[Al(H_2O)_6]^{3+} \longrightarrow [Al(OH)_2(H_2O)_4]^{+}$
- $MnO_4^{2-} \longrightarrow MnO_4^{-}$
- $[CuCl_4]^{2-} \longrightarrow [CuCl_2]^{-}$

Part A adapted with permission from UCLES, A-Level Chemistry, June 1992, Paper 4, Question 15;
Part B adapted with permission from OCSEB, A-Level Chemistry, June 1994, Paper 1, Question 12

Question deck:

STEM SMART Chemistry Week 10

Oxidation Numbers of Halides

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Part A

Oxidation of bromine

What changes can be regarded as oxidation of bromine?

1. $Br_2 \longrightarrow BrO^-$
2. $Br_2 \longrightarrow BrF$
3. $Br_2 \longrightarrow BrI$

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct

Part B

Oxidation numbers of halides

Which of the statements about the reaction below are correct?

$$IO_3^-(aq) + 2\,I^-(aq) + 6\,H^+(aq) + 6\,Cl^-(aq) \longrightarrow 3\,ICl_2^-(aq) + 3\,H_2O(l)$$

1. The oxidation number of chlorine changes from -1 to -2.
2. The oxidation number of the iodine in the iodide ion $I^-(aq)$ changes from -1 to $+1$.
3. The oxidation number of the iodine in the iodate ion $IO_3^-(aq)$ changes from $+5$ to $+1$.

- **1, 2** and **3** are correct
- **1** and **2** only are correct
- **2** and **3** only are correct
- **1** only is correct

Part A adapted with permission from UCLES, A-Level Chemistry, June 1996, Paper 3, Question 37;
Part B adapted with permission from UCLES, A-Level Chemistry, June 1989, Paper 3, Question 37

Question deck:
STEM SMART Chemistry Week 10

Platinum, Hydroquinone and Silver

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level P1

Part A

Oxidation numbers of platinum

The anti-cancer drug *cis-platin* has the formula $Pt(NH_3)_2Cl_2$.

In the human body, one of the chloride ions of *cis-platin* is replaced by one water molecule.

$$Pt(NH_3)_2Cl_2 + H_2O \longrightarrow [Pt(NH_3)_2(H_2O)Cl]^+ + Cl^-$$

What is the oxidation number of platinum in each of these complexes?

	cis-platin	*in the aquo complex*
A	$+2$	$+1$
B	$+2$	$+2$
C	$+4$	$+3$
D	$+4$	$+4$

- A
- B
- C
- D

Part B

Hydroquinone and silver bromide

When exposed film from a camera is developed, one step involves reacting the light-activated silver bromide crystals with hydroquinone in alkali according to the following equation:

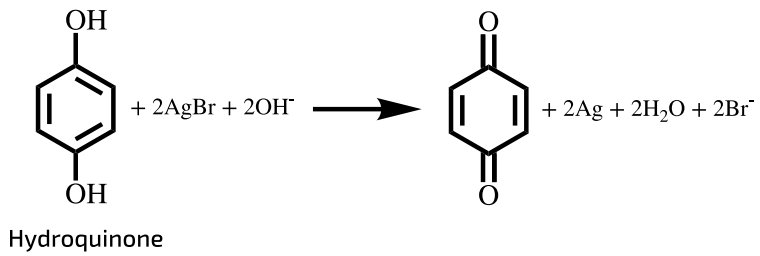

Figure 1: Hydroquinone and silver bromide in presence of base.

Which of the following describes the role of hydroquinone?

- It acts only as an acid
- It acts only as an oxidising agent
- It acts as both an acid and a reducing agent
- It acts as both a base and a reducing agent
- It acts only as a reducing agent

Part A adapted with permission from UCLES, A-Level Chemistry, June 1996, Paper 3, Question 9;
Part B adapted with permission from OCSEB, A-Level Chemistry, June 1994, Paper 1, Question 3

Question deck:
STEM SMART Chemistry Week 10

Ferrite

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C1

Aerials in portable radios are made of a mixed oxide of calcium and iron known as 'Ferrite'. It contains $18.5\,\%$ calcium and $51.9\,\%$ iron by mass. Calculate the empirical formula of 'Ferrite' and hence deduce the oxidation number of the iron it contains.

Part A

Empirical formula

Empirical Formula:

Part B

Oxidation number

Oxidation number:

Adapted with permission from UCLES, A-Level Chemistry, June 1992, Paper 2, Question 3

Question deck:

STEM SMART Chemistry Week 10

Oxides of Nitrogen

Subject & topics: Chemistry | Inorganic | Redox **Stage & difficulty:** A Level C2

Part A

Oxides of nitrogen

In an attempt to establish the formula of an oxide of nitrogen, a known volume of the pure gas was mixed with hydrogen and passed over a catalyst at a suitable temperature. $100\,\%$ conversion of the oxide to ammonia and water was shown to have taken place.

$$\mathrm{N}_x\mathrm{O}_y \xrightarrow[\text{catalyst}]{\mathrm{H}_2} x\,\mathrm{NH}_3 + y\,\mathrm{H}_2\mathrm{O}$$

$2400\,\mathrm{cm}^3$ of the nitrogen oxide, measured at room temperature and pressure (RTP) produced $7.200\,\mathrm{g}$ of water. The ammonia produced was neutralised by $200\,\mathrm{cm}^3$ of $1.0\,\mathrm{mol\ dm}^{-3}$ HCl.

[Molar volume of gas at RTP = $24000\,\mathrm{cm}^3\ \mathrm{mol}^{-1}$]

What was the oxidation number of nitrogen in the solid oxide?

- +1
- +2
- +3
- +4
- +5

Part B

Oxidation numbers of nitrogen

The key stage in the manufacture of nitric acid is the reaction of ammonia with air in the presence of a platinum-rhodium gauze:

$$4\,NH_3\,(g) + 5\,O_2\,(g) \longrightarrow 4\,NO\,(g) + 6\,H_2O\,(g)$$

What is the oxidation number of nitrogen in

NH_3

NO

Part A adapted with permission from UCLES, A-Level Chemistry, November 1989, Paper 3, Question 2;
Part B adapted with permission from UCLES, A-Level Chemistry, November 1995, Paper 3, Question 1

Question deck:
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